Linear Algebra | Test for Chapter 4 | November 6, 2026

Name: _____ R. Hammack Score: _____

**Directions:** Put away phones, calculators, smart watches, notes, etc. Show work & answer in the space provided.

1. Write $\mathbf{v} = (10, 1, 4)$ as a linear combination of $\mathbf{u}_1 = (2, 3, 5)$, $\mathbf{u}_2 = (1, 2, 4)$ and $\mathbf{u}_3 = (-2, 2, 3)$.

2. Consider the subset $W = \{f \in P_4 \ : \ f(1) = 0\} \subseteq P_4$. Is $W$ a subspace of $P_4$? Explain.

3. Find a basis for the nullspace of the matrix $A = \begin{bmatrix} 1 & 3 & -2 & 4 \\ 0 & 1 & -1 & 2 \\ -2 & -6 & 4 & -8 \end{bmatrix}$.

4. Suppose the set $S = \{\mathbf{v}, \mathbf{w}\}$ is linearly independent. Without knowing any further information, is it possible to say whether or not the set $S' = \{\mathbf{v}+\mathbf{w}, \ \mathbf{v}-\mathbf{w}\}$ is linearly independent or dependent? Explain.

5. Suppose a matrix $A$ has 45 columns and its rows are linearly independent.
Also, the set $\{\mathbf{x} \in \mathbb{R}^{45} \mid A\mathbf{x} = \mathbf{0}\}$ is 7-dimensional. How many rows does $A$ have? Explain.

6. The set $B = \{1 + x + x^2,\ 2 + x + x^2,\ 1 - x + x^2\}$ is a basis for $P_2$. Consider $3 + x^2 \in P_2$.
Find the coordinate vector $[3 + x^2]_B$.